higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIALS
NQF LEVEL 4

13 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 1.1.1 Thermosetting
1.1.2 Heated
1.1.3 Polymer
1.1.4 Adhesives
1.1.5 Sap or latex
1.1.6 PVC

(6 × 1) (6)

1.2
- Clean the surfaces that must be sealed of all dirt particles, rust and previous sealing agents.
- If the manufacturer specifies a primer, apply the specified primer to the cleaned surfaces.
- Using a caulking gun, apply the sealant and force it into the joint in the masonry or concrete.
- Leave the sealant to cure and harden. (4)

1.3
- Cut a small slither of material with a sharp knife.
- If it cuts cleanly, it is a thermoplastic.
- If it crumbles, it is a thermoset. (3)

1.4
- Aramid fibres
- Carbon fibres
- Glass fibres (3)

1.5
- The finished product can be delivered to the site, ready for installation.
- Composites are easy to make and install.
- They are resistant to heat and chemicals.
- They are cost effective. (4)

[20]

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QUESTION 2

2.1
- To make bridge bearing
- To waterproof ceiling and roofs
- As bonding
- To absorb shock
- As plumbing rubbers and seals (5)

2.2 The primer is a paint-like material one applies to the clean and dry surface of a joint to improve the bond between the surface of the joint and sealant that is used. (2)

2.3
- Wood impregnated with oil
- Copper, zinc, steel or brass
- Teflon-coated materials
- Concrete or masonry (Any 3 × 1) (3)

[10]

QUESTION 3

3.1
- They are not resistant to many chemicals.
- Their curing time varies greatly between materials.
- They cannot withstand a large temperature range.
- Their anaerobic curing process makes them brittle.
- They can be very harmful to the skin and eyes.
- They shrink when they cure. (6)

3.2
- Clean both surfaces thoroughly to get rid of any dirt oil or grease.
- Try to roughen both surfaces because adhesives will stick better to rough surfaces.
- Brush off any excess loose dirt.
- Apply the adhesive to both surfaces.
- Allow both surfaces to become touch dry (6–10 minutes).
- When satisfied, align the two surfaces.
- Apply an equal amount of pressure to both surfaces until they are securely bonded. (7)

3.3
- Can cause allergies, dermatitis and breathing problems.
- Epoxies have limited resistance to UV light.
- Difficult to mix and prone to waste during mixing process.
- They have limited resistance to most organic acids.
- Cannot be mixed and cured in cold temperatures. (Any 2 × 1) (2)

[15]

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QUESTION 4

4.1
- High blood pressure
- Damage to the brain and nervous system
- Digestive problems
- Loss of memory
- Reduced concentration
- Muscle and joint pain (Any 5 × 1) (5)

4.2
- Sand the surface gently to remove splinters and small imperfections.
- Wipe the surface with a damp cloth to remove dust or grit.
- Seal all knots with knotting sealer.
- Apply a quick drying oil or latex-based primer to protect against water absorption.
- Fill up all holes with an appropriate filling compound.
- Apply primer and wait for it to dry.
- Sand it down and apply the undercoat.
- Apply the finishing coat. (8)

4.3
- Primer/base coat
- Undercoating/intermediate coating
- Top coat/final coat (3)

4.4
- Etch the substrata with a mild acid.
- Immediately clean the acid off the substrate with fresh water.
- Apply the primer with a brush.
- Apply one or more solvent-based finishing coats over the primer. (4)

[20]

QUESTION 5

5.1 Efflorescence is powdery white salt or potassium is left on masonry or concrete surface as it dries out. (2)

5.2
- Silicones
- Organo-silanes
- Silanes (3)

5.3 5.3.1 Footings – that part of a construction designed to transfer loads to the supporting foundation, usually constructed of reinforced concrete to support base brickwork

5.3.2 DPC – a continuous layer of impervious material placed in a masonry wall or between a floor and wall to prevent the upward or downward migration of moisture

5.3.3 Weep-hole – Narrow joint (hole) in masonry wall left open to allow water from behind the wall to escape

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5.3.4 Flashing – Pieces of sheet metal or roll roofing fitted to the joint of any intersection, penetration or projection to prevent water from penetrating the structure

5.3.5 Cavity wall – A wall that consists of two parallel single-leaf masonry walls separated by a gap or cavity

(5 × 2) (10)
[15]

QUESTION 6

6.1
- Induction sessions
- Risk assessment
- Toolbox meetings
- Training
- Protective clothing (Any 4 × 1) (4)

6.2
- Economical to manufacture and to use
- Provides a durable surface finish
- Highly resistant to damage by the elements
- Resistance to damage by heavy traffic and heavy loads
- Easy to apply by spreading and compacting (Any 4 × 1) (4)

6.3 6.3.1 Scraper
6.3.2 Smooth drum vibrating roller

(2 × 1) (2)
[10]

QUESTION 7

7.1
- Countersunk screw
- Nylon-insert lock nut
- Anchor bolt (3)

7.2
- To maintain quality control
- To reduce the possibility of wrong bolt application
- To provide torque or force indication specification
- To specify the tensioning required in certain sections of a structure

(Any 3 × 1) (3)

7.3
- There could be oil present between the contact surfaces.
- There could be rust present between the contact surfaces.
- Tolerances in the assembly of the structure
- Bearing friction between the contact surfaces
- Friction between the bolt head and the nut thread (Any 4 × 1) (4)

[10]

TOTAL: 100

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